

Arjuna JEE 2025

Practice Sheet (Mathematics)

Permutation and Combination

- Q1** The number of 10 digit numbers that can be formed by using the digits 2 and 3 is
 (A) $^{10}C_2 + ^9C_2$
 (B) 2^{10}
 (C) $2^{10} - 2$
 (D) $10!$
- Q2** If P_r stands for rP_r , then sum of the series $1 + P_1 + 2P_2 + 3P_3 + \dots + nP_n$ is
 (A) P_{n+1}
 (B) $P_{n+1} - 1$
 (C) $P_{n-1} + 1$
 (D) None of these
- Q3** Rakshit is allowed to select $(n+1)$ or more books out of $(2n+1)$ distinct books. If the number of ways in which he may not select all of them is 63, then value of n is
 (A) 3 (B) 4
 (C) 5 (D) 11
- Q4** The number of ways in which a committee of 3 women and 4 men be chosen from 8 women and 7 men if Ms. X refuses to serve on the committee if Mr. Y is member of the committee is
 (A) 420 (B) 840
 (C) 1540 (D) 720
- Q5** The product of first n odd natural numbers equals
 (A) $({}^{2n}C_n) ({}^nP_n)$
 (B) $(\frac{1}{2})^n ({}^{2n}C_n) ({}^nP_n)$
 (C) $(\frac{1}{4})^n ({}^{2n}C_n) ({}^nP_n)$
 (D) $\frac{1}{n!} ({}^{2n}C_n)$
- Q6** The number of integral solutions of $x + y + z = 0$ with $x \geq -5, y \geq -5, z \geq -5$ is
 (A) 134 (B) 136
 (C) 138 (D) 140
- Q7** The number of six-digit numbers which have sum of their digits as an odd integer, is
 (A) 45000 (B) 450000
 (C) 97000 (D) 970000
- Q8** Ten different letters of an alphabet are given. Words with five letters are formed from these given letters. The number of words which have at least one of their letters repeated is
 (A) 69760 (B) 30240
 (C) 99748 (D) 60480
- Q9** The number of ways in which we can select four numbers from 1 to 30 so as to exclude every selection of four consecutive numbers is
 (A) 27378 (B) 27405
 (C) 27397 (D) 19050
- Q10** If ${}^nC_4, {}^nC_5$ and nC_6 are in A.P., the value of n can be
 (A) 14 (B) 11
 (C) 9 (D) 5
- Q11** The letters of the word SURITI are written in all possible orders and these words are written out as in a dictionary. Then the rank of the word SURITI is
 (A) 236 (B) 245
 (C) 307 (D) 315
- Q12** A candidate is required to answer 6 out of 10 questions which are divided into two groups, each containing 5 questions. He is not permitted to attempt more than 4 questions from either group. The number of different ways



in which the candidate can choose six question is

- (A) 50 (B) 150
(C) 200 (D) 250

Q13 Consider the number $N = 2910600$. On the basis of above information, answer the following questions: Total number of divisors of N , which are divisible by 15 but not by 36 are-

- (A) 92 (B) 94
(C) 96 (D) 98

Q14 Consider the number $N = 2910600$. On the basis of above information, answer the following

Total number of ways, in which the given number can be split into two factors such that their highest common factor is a prime number is equal to-

- (A) 16 (B) 32
(C) 48 (D) 64

Q15 The value of n if

Column-I		Column-II	
(A)	${}^{n+4}C_{n+1} - {}^{n+3}C_n = 15(n+2)$	(P)	6
(B)	$(11)^nP_4 = (20)({}^{n-2}P_4)$	(Q)	10
(C)	${}^{2n}C_3 = 11^n C_3$	(R)	16
(D)	${}^{14}C_{n-2}, {}^{14}C_{n-1}, {}^{14}C_n$ are in A.P.	(S)	27

- (A) $A \rightarrow S; B \rightarrow R; C \rightarrow P; D \rightarrow P, Q$
(B) $A \rightarrow R; B \rightarrow S; C \rightarrow P; D \rightarrow Q$
(C) $A \rightarrow R; B \rightarrow P; C \rightarrow Q; D \rightarrow S$
(D) $A \rightarrow S; B \rightarrow R; C \rightarrow P; D \rightarrow Q$

Q16 Consider all possible permutations of the letters of the word ENDEANOEL.

Match the statement/Expressions on the left with the Statements/Expressions on the right

Column-I		Column-II	
(A)	The number of permutations containing the word ENDEA is	(P)	$5!$
(B)	The number of permutations in which the letter E occurs in the first and the last position is	(Q)	$2 \times 5!$
(C)	The number of permutations in which none of the letters D, L, N occurs in the last five positions is	(R)	$7 \times 5!$
(D)	The number of permutations in which the letters A, E, O occur only in odd position is	(S)	$21 \times 5!$

- (A) $A \rightarrow P; B \rightarrow Q; C \rightarrow R; D \rightarrow S$
(B) $A \rightarrow P; B \rightarrow S; C \rightarrow Q; D \rightarrow Q$
(C) $A \rightarrow R; B \rightarrow P; C \rightarrow Q; D \rightarrow S$
(D) $A \rightarrow S; B \rightarrow R; C \rightarrow P; D \rightarrow Q$

Q17 If m = the number of three digit number of the form xyz with $x, z < y, x \neq 0$, then $\frac{m}{120}$ is

Q18 If m = the number of natural numbers lying between 100 and 1000 that can be formed by using digits 1, 2, 3, 5, 7 and 0 when repetition of digit is allowed then $m - 171$ is

Q19 A straight is a five card hand containing consecutive values. If m = the number of ways in which all the five cards are not from the same suit, then $m/4590$ is

Q20 A 5-digit number divisible by 3 is to be formed using the digits 0, 1, 2, 3, 4 and 5 without repetition. The total number of ways in which this can be done is

- (A) 216 (B) 600
(C) 240 (D) 3125

Q21 If all the words formed from the letters of the word HORROR are arranged in the opposite order as they are in a dictionary, then the rank of the word HORROR is.

- (A) 56 (B) 57



(C) 58

(D) 59

- Q22** A person goes for an examination in which there are four papers with a maximum of m marks from each paper. The number of ways in which one can get $2m$ marks, is

(A) $^{2m+3}C_3$
 (B) $\left(\frac{1}{3}\right)(m+1)(2m^2+4m+1)$
 (C) $\left(\frac{1}{3}\right)(m+1)(2m^2+4m+3)$
 (D) None of these

- Q23** The letters of the word KANPUR are arranged in all possible ways as in a dictionary, the rank of the word KANPUR from last is

(A) 598 (B) 599
 (C) 600 (D) 601

- Q24** A man has three friends. The number of ways he can invite one friend everyday for dinner on six successive nights, so that no friend is invited more than three times, is

(A) 360 (B) 420
 (C) 170 (D) 510

- Q25** The number of ways in which the letters of the word PERSON can be placed in the squares of the given figure so that no row remains empty is



(A) $24 \times 6!$
 (B) $26 \times 6!$
 (C) $26 \times 7!$
 (D) $27 \times 6!$

- Q26** There are 10 points in a plane of which no three are collinear and 4 points are concyclic the number of different circles that can be drawn through atleast 3 of these, is

(A) 116 (B) 120
 (C) 117 (D) none

- Q27** The number of all five digit numbers which are divisible by 4 that can be formed from the digits

0, 1, 2, 3, 4 (without repetition), is

(A) 36 (B) 30
 (C) 34 (D) None of these

- Q28** Let P_n denotes the number of ways in which three people can be selected out of n people sitting in a row, if no two of them are consecutive. If, $P_{n+1} - P_n = 15$ then the value of n is ____.

- Q29** The total number of six-digit natural numbers that can be made with the digits 1, 2, 3, 4, if all digits are to appear in the same number at least once is?

(A) 1560 (B) 840
 (C) 1080 (D) 480

- Q30** In how many ways can one select a cricket team of eleven from 17 players in which only 5 persons can bowl if each cricket team of 11 must include exactly 4 bowlers?

- Q31** Total number of 6—digit numbers in which only and all the five digits 1, 3, 5, 7 and 9 appear, is

(A) $\frac{5}{2}(6!)$
 (B) $6!$
 (C) 5^6
 (D) $\frac{1}{2}(6!)$

- Q32** The set $S = \{1, 2, 3, \dots, 12\}$ to be partitioned into three sets

A, B, C of equal size. Thus,
 $A \cup B \cup C = S, A \cap B = B \cap C = A \cap C = \phi$

The number of ways to partition S , is

(A) $\frac{12!}{3!(4!)^3}$
 (B) $\frac{12!}{3!(3!)^4}$
 (C) $\frac{12!}{(4!)^3}$
 (D) $\frac{12!}{(3!)^4}$

- Q33** If the four-letter words (need not be meaningful) are to be formed using the letters from the word "MEDITERRANEAN" such that the



first letter is **R** and the fourth letter is **E**, then the total number of such words, is

- (A) $\frac{11!}{(2!)^3}$
 (B) 59
 (C) 110
 (D) 56

Q34 What is sum of all rearrangements of the 4-digit number 3321?

- (A) 29999
 (B) 27777
 (C) 28888
 (D) 29997

Q35 If m parallel lines in a plane are intersected by a family of n parallel lines, the number of parallelograms that can be formed is?

- (A) $\frac{1}{4}mn(m-1)(n-1)$
 (B) $\frac{1}{2}mn(m-1)(n-1)$
 (C) $\frac{1}{4}m^2n^2$
 (D) none of these

Q36 There are 3 set of parallel lines containing respectively p lines, q lines and r lines respectively. The greatest number of parallelograms that can be formed by the system

- (A) $pqr + (p-1)(q-1)(r-1)$
 (B) $\frac{1}{4}\{pqr + (p-1)(q-1)(r-1)\}$
 (C) $\frac{1}{4}pqr(p+1)(q+1)(r+1)$
 (D) $({}^pC_2)({}^qC_2) + ({}^qC_2)({}^rC_2) + ({}^rC_2)({}^pC_2)$

Q37 The number of ways in which 10 candidates $A_1, A_2, \dots, A_{10}$ can be ranked such that A_1 is always above A_{10} is

- (A) 5!
 (B) $2(5!)$
 (C) 10!
 (D) $\frac{1}{2}(10!)$

Q38 If a, b , and c are positive integers such that $a + b + c \leq 8$, the number of possible values of the ordered triplet (a, b, c) is

- (A) 84
 (B) 56
 (C) 83
 (D) None of these

Q39 How many different words can be formed by Jumbling the letters in the word MISSISSIPPI in which no two Ss are adjacent?

- (A) $6 \cdot 7 \cdot {}^8C_4$
 (B) $6 \cdot 8 \cdot {}^7C_4$
 (C) $7 \cdot {}^6C_4 \cdot {}^8C_4$
 (D) $8 \cdot {}^6C_4 \cdot {}^7C_4$

Q40 Number of ways in which 25 identical things be distributed among five persons if each gets odd number of things is

- (A) $25C_4$
 (B) $12C_8$
 (C) $14C_{10}$
 (D) $13C_3$

Q41 Let m = number of ways in which letters of the word KIAAN can be permuted so that both the consonants are together and n = number of ways in which letters of word MIHIR can be permuted so that two I's are together, then n/m is

Q42 20 persons were invited for a party. The number of ways they and the host be seated at a circular table if two particular persons sit on either side of the host, is

- (A) 18!
 (B) $18! \times 2!$
 (C) $18!/2!$
 (D) 19!

Q43 The number of ways of choosing a committee of two women and three men from five women and six men, if Mr. A refuses to serve on the committee if Mr. B is a member and Mr. B can only serve, if Ms. C is the member of the committee is?

- (A) 60
 (B) 84
 (C) 124
 (D) none of these

Q44 In a row, there are 81 rooms, whose door no. are 1, 2, ..., 81, initially all the door are closed. A person takes 81 round of the row, numbers as 1st round, 2nd round 81th round. In each



round, he interchange the position of those door number, whose number is multiple of the round number. Find out after 81^{st} round, How many doors will be open.

- Q45** Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is?
- (A) 264
(B) 265
(C) 53
(D) 67

- Q46** The number of groups that can be made from 5 different green balls, 4 different blue balls and 3 different red balls, if atleast one green and one blue ball is to be included
- (A) 3700 (B) 3720
(C) 4340 (D) None of these

- Q47** Sixteen players $S_1, S_2, S_3, \dots, S_{16}$ play in a tournament. Number of ways in which they can be grouped into eight pairs so that S_1 and S_2 are in different pairs, is?
- (A) $\frac{(14)!}{2^6 \cdot 6!}$
(B) $\frac{(15)!}{2^7 \cdot 7!}$
(C) $\frac{(14)!}{2^7 \cdot 6!}$
(D) $\frac{(14)!}{2^6 \cdot 7!}$

- Q48** In a room, there are 12 bulbs of the same wattage, each having a separate switch. The number of ways to light the room with different amount of illumination is.
- (A) $12^2 - 1$
(B) 2^{12}
(C) $2^{12} - 1$
(D) 12^2

Q49

We have to choose 11 players for cricket team from eight batsmen, six bowlers, four all-rounders, and two wicket keepers in the following conditions. The number of selections when a particular batsman and a particular wicket keeper do not want to play together is?

- (A) $2^{18} C_{10}$
(B) ${}^{19}C_{11} + {}^{18}C_{10}$
(C) ${}^{19}C_{10} + {}^{19}C_{11}$
(D) none of these

- Q50** Let 52 cards of a deck be arranged in a line. Number of ways in which spades appear in increasing order of denomination is?

- (A) $\frac{52!}{13!}$
(B) ${}^{52}C_{13} \times 2^{39}$
(C) ${}^{52}C_{13} \times 13!$
(D) $\frac{52!}{13!13!}$



Answer Key

Q1 (B)
Q2 (A)
Q3 (A)
Q4 (C)
Q5 (B)
Q6 (B)
Q7 (B)
Q8 (A)
Q9 (A)
Q10 (A)
Q11 (A)
Q12 (C)
Q13 (C)
Q14 (C)
Q15 (A)
Q16 (B)
Q17 2
Q18 8
Q19 2
Q20 (A)
Q21 (C)
Q22 (C)
Q23 (B)
Q24 (D)
Q25 (B)

Q26 (C)
Q27 (B)
Q28 8
Q29 (A)
Q30 3960
Q31 (A)
Q32 (C)
Q33 (B)
Q34 (D)
Q35 (A)
Q36 (D)
Q37 (D)
Q38 (B)
Q39 (C)
Q40 (C)
Q41 1
Q42 (B)
Q43 (C)
Q44 9
Q45 (C)
Q46 (B)
Q47 (A)
Q48 (C)
Q49 (B)
Q50 (A)



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